HOMEWORK 5.2 · UNIT 5 · NC.8.G.6 · NC.8.G.7

Pythagorean Theorem Day 2

Jan 6 / Jan 7 · Targets: G.6-4, G.7-3, G.6-7, G.6-6, G.7-5, G.6-9, G.6-8

Name _____ Date _____ Block _____

Remember: Converse: if $a^2 + b^2 = c^2$ (c = the longest side), the triangle **is** a right triangle. If the two sides are not equal, it is **not**. Example: 6, 8, 10 → 36 + 64 = 100 and $10^2 = 100$ ✓ right triangle.

SCORE 3.0 Approaching · the basics

Targets: G.6-4, G.7-3

1 Is 12, 16, 20 a right triangle? G.6-4

Yes No

Why? _____

2 Is 5, 7, 9 a right triangle? G.6-4

Yes No

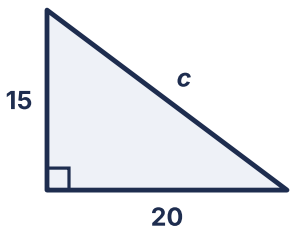
Why? _____

3 Which side lengths make a right triangle? G.6-4

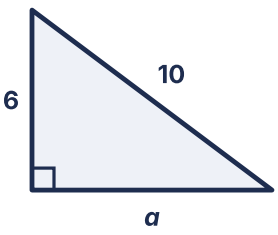
☐ 8, 15, 17 ☐ 4, 5, 6 ☐ 10, 24, 26 ☐ 7, 24, 26
☐ 1, 1, 2

Select all that apply.

4 Find c. REVIEW 5.1 G.7-3



5 Find the missing leg a. REVIEW 5.1 G.7-3



SCORE 4.0 Proficient · on target

Targets: G.6-7, G.6-6, G.7-5

6 **Kim** made a mistake. Circle it. Explain it. Fix it. G.6-7

KIM'S WORK

Sides 9, 40, 41:

$9 + 40 = 49$

$49 \neq 41$, so it is not a right triangle.

7 A garden bed has sides 5 ft and 12 ft. The diagonal is 13 ft. Is the corner a right angle? Prove it. G.6-6

8 A TV is 18 in tall and 24 in wide. How long is the diagonal? REVIEW 5.1 G.7-5

Ⓐ 30 in Ⓑ 42 in Ⓒ 900 in Ⓓ 6 in

SCORE 5.0 Mastery · go deeper

Targets: G.6-9, G.6-8

9 Which tool fits each job: the theorem or the converse? (A) A ramp rises 1 ft over a flat distance of 12 ft. How long is the ramp? (B) A wall corner measures 6 ft, 8 ft, 10 ft. Is it square? G.6-9

(A) uses the _____ because _____ . (B) uses the _____ because _____ .

10 A triangle has sides 6, 8 and 11. Is it a right triangle? If not, how long would the longest side need to be? G.6-8

_____ , because _____ . It would need to be _____ .